

# Fundamentals of Astronomy

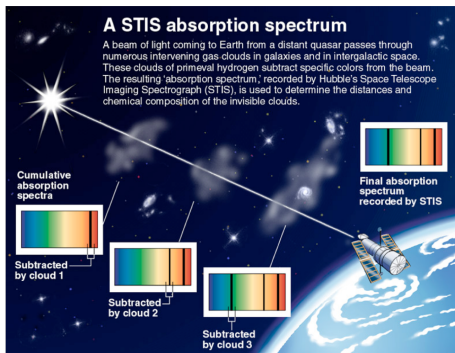
## Radiation and Matter

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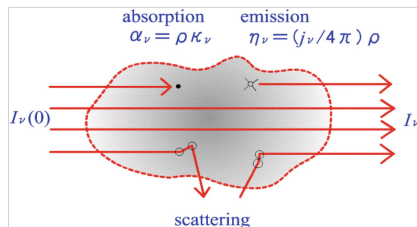
# Introduction I

Electromagnetic radiation emitted by a source propagates through space where it can be modified by the interaction with intervening matter. Galactic hot gases or dust clouds can cool by emitting some amounts of radiation. The radiation that is emitted in one region can be absorbed by other matter inside or outside the region.



# Introduction II

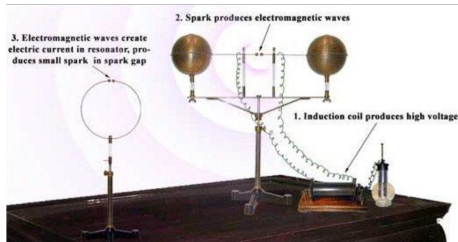
This is typically the kind of radiation that we observe with our telescopes. Therefore to understand our observations we need to know the basic mechanisms of emission, absorption and transport of radiation.



First, we will describe the electromagnetic radiation in terms of electromagnetic waves and photons.

# Maxwell's Equations I

Maxwell (1831-1879), introduces the concept that electricity and magnetism are actually two components of the same physical phenomenon and realise that optics is a subfield of the (Classical) Electrodynamics. This is summarized in Maxwell's Equations that are four of the most famous and important equations in science. Maxwell showed that light is a form of electromagnetic radiation as was later confirmed experimentally by Hendrich Hertz in 1888.



# Maxwell's Equations II

Although, Maxwell's Equations were formulated to describe the electrical and magnetic fields and their interactions caused by **macroscopic distributions** of electric charges and currents, early in the 20th century, Hendrick Antoon Lorentz took the electrodynamics theory further to the **microscopic scale**.

The four Maxwell's equations are:

$$\nabla \cdot \mathbf{E}(t, \mathbf{r}) = \frac{\rho(t, \mathbf{r})}{\epsilon_0} \quad (1)$$

$$\nabla \times \mathbf{E}(t, \mathbf{r}) = -\frac{\partial \mathbf{B}(t, \mathbf{r})}{\partial t} \quad (2)$$

$$\nabla \cdot \mathbf{B}(t, \mathbf{r}) = 0 \quad (3)$$

$$\nabla \times \mathbf{B}(t, \mathbf{r}) = \mu_0 \mathbf{J}(t, \mathbf{r}) \quad (4)$$

It is to note that in these equations there is an asymmetry: while a varying magnetic field produces an electric field, the converse is not true.

Maxwell noticed this and introduced an extra term, the "displacement current", which behaves like a current density flowing in vacuum. The existence of this term

# Maxwell's Equations III

has the physical consequences of the existence of electromagnetic radiation that can carry energy and momentum over very long distances, even in vacuum.

Maxwell published his *Treatise on Electricity and Magnetism* in 1873. The theory was recast into the modern vector-calculus form of *Maxwell's Equations* by Oliver Heaviside in 1884:

$$\nabla \cdot \mathbf{E}(t, \mathbf{r}) = \frac{\rho(t, \mathbf{r})}{\epsilon_0} \quad (5)$$

$$\nabla \times \mathbf{E}(t, \mathbf{r}) = -\frac{\partial \mathbf{B}(t, \mathbf{r})}{\partial t} \quad (6)$$

$$\nabla \cdot \mathbf{B}(t, \mathbf{r}) = 0 \quad (7)$$

$$(8)$$

$$\nabla \times \mathbf{B}(t, \mathbf{r}) = \mu_0 \mathbf{J}(t, \mathbf{r}) + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}(t, \mathbf{r})}{\partial t} \quad (9)$$

# Maxwell's Equations IV

The first equation is the **Gauss's Law**; the second is the **Faraday-Maxwell law**, the third express the non-existence of magnetic monopoles, the last is the **Ampere-Maxwell law**.

Another name often used for this set of equations is the **Maxwell-Lorentz** equations.

The constant  $\epsilon_0 = \frac{10^7}{4\pi c^2} \approx 8.8542 \cdot 10^{-12} \left( \frac{C^2}{m^2 \cdot N} \right)$  is the *vacuum permittivity* and  $c \approx 2.9979108 \text{ m/s}$  is the speed of light in vacuum.

The constant  $\mu_0 = 4\pi 10^{-7} \text{ H/m}$  is the *magnetic permeability*. It is to note that:  $\mu_0 \epsilon_0 = \frac{1}{c^2}$ .

Given  $\rho$  and  $\mathbf{J}$ , the initial and boundary conditions, the Maxwell's equations form a system of well-posed partial differential equations which allow to completely determine  $\mathbf{E}$  and  $\mathbf{B}$ .

# The conservation of the charge I

Taking the divergence of equation 9 we have:

$$\nabla \cdot (\nabla \times \mathbf{B}(t, \mathbf{r})) = \mu_0 \nabla \cdot \mathbf{J}(t, \mathbf{r}) + \mu_0 \epsilon_0 \frac{\partial \nabla \cdot \mathbf{E}(t, \mathbf{r})}{\partial t} \quad (10)$$

then, because for any vector field  $\mathbf{a}$ ,  $\nabla \cdot \nabla \times \mathbf{a} = 0$ , and from the Gauss's law, we have

$$\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0 \quad (11)$$

This is a continuity equations that express the local "conservation of the charge". In the vacuum, its meaning is that the charge density varies in each point of the space in which the divergence of the current density is different from zero.

That is, if there is a reduction of the charge in a region of volume  $V$  this reduction must be equal to the density of current through the surface delimiting  $V$ .



# The Lorentz Force I

Suppose we have some charge and current configuration which, at time  $t$ , produces the fields  $\mathbf{E}$  and  $\mathbf{B}$ , the effect of  $\mathbf{E}$  and  $\mathbf{B}$  on an elementary charge  $e$  moving with a velocity  $\mathbf{v}$  is given by the *Lorentz's force*:

$$\mathbf{F} = e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (12)$$

This allows an operational definition of  $\mathbf{E}$  and  $\mathbf{B}$ .

After a time  $dt$  the charge will move a bit then the work for unit time done by the fields on a charged particle is:

$$\mathbf{v} \cdot \mathbf{F} = e(\mathbf{v} \cdot \mathbf{E}) \quad (13)$$

Since  $\mathbf{F} = m \frac{d\mathbf{v}}{dt}$  we have:

$$\mathbf{v} \cdot \mathbf{F} = \frac{d}{dt} \left( \frac{1}{2} m v^2 \right) = \frac{dU_{mec}}{dt} = e(\mathbf{v} \cdot \mathbf{E}) \quad (14)$$

# The Lorentz Force II

This result can be generalized to the total force on a distribution of charges  $\rho$  contained in unit volume.

The Lorentz's force for unit of volume is ( $\mathbf{J} = \rho\mathbf{v}$ ):

$$\mathbf{f} = \rho\mathbf{E} + \mathbf{J} \times \mathbf{B} \quad [N/m^3] \quad (15)$$

Then the work for unit time done by the field on a unit volume is:

$$\frac{du_{mec}}{dt} = \mathbf{J} \cdot \mathbf{E} \quad [W/m^3] \quad (16)$$

This equation is known as Joule heating; it expresses the rate of energy transfer to the charge carriers from the fields. It is the microscopic version of the famous  $P = VI$ . This term only contains the electric field because the magnetic field can do no work on the charges.

# The Poynting's vector I

Energy conservation is a physical principle that also extends to electromagnetic phenomena. If a system is subject to the action of an electromagnetic field, therefore, energy transfers can occur between the field and the system, as long as the sum of all forms of energy possessed by the system and the interacting field remains constant.

The term  $\mathbf{J} \cdot \mathbf{E}$  in equation (16), can be expressed in function of the fields using the Ampère–Maxwell law (9):

$$\mathbf{J} \cdot \mathbf{E} = \frac{1}{\mu_0} \left[ \mathbf{E} \cdot (\nabla \times \mathbf{B}) - \epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} \right] \quad (17)$$

with some algebra and standard vector calculus rules we have:

$$\mathbf{J} \cdot \mathbf{E} + \frac{1}{2} \frac{\partial}{\partial t} \left( \epsilon_0 E^2 + \frac{B^2}{\mu_0} \right) = -\frac{1}{\mu_0} \nabla \cdot (\mathbf{E} \times \mathbf{B}) \quad (18)$$

This is the **Poynting's Theorem**.

# The Poynting's vector II

The second term on the left side of (18):

$$u_{em} = u_E + u_B = \frac{1}{2}(\epsilon_0 E^2 + \frac{B^2}{\mu_0}) \quad (19)$$

is the **total energy density** stored in the electromagnetic field.

Integrating over the volume  $V$  of interest:

$$\iiint_V (\mathbf{J} \cdot \mathbf{E}) dV + \frac{\partial}{\partial t} \iiint_V u_{em} dV = -\frac{1}{\mu_0} \iiint_V \nabla \cdot (\mathbf{E} \times \mathbf{B}) dV$$

if  $S$  is the surface delimiting  $V$ , using the divergence theorem:

$$\iiint_V (\mathbf{J} \cdot \mathbf{E}) dV + \frac{\partial}{\partial t} \iiint_V u_{em} dV = -\frac{1}{\mu_0} \iint_S (\mathbf{E} \times \mathbf{B}) \cdot d\mathbf{S}$$

# The Poynting's vector III

The quantity:

$$\mathbf{S} = \frac{1}{\mu_0}(\mathbf{E} \times \mathbf{B}) = \mathbf{E} \times \mathbf{H} \quad [W/m^2] \quad (20)$$

$$\mathbf{H} = \mathbf{B}/\mu_0.$$

is the energy flux density, that is the energy per unit time, per unit area transported by the fields through the surface  $S$  is called the **Poynting vector**: Therefore, The integral form of the Poynting's Theorem over a volume  $V$  with bounding surface  $S$

$$\frac{d}{dt}(u_{mec} + u_{em}) = - \iint_S \mathbf{S} \cdot d\mathbf{S} \quad (21)$$

can be interpreted saying that the rate of change of the mechanical energy for unit volume plus the rate of change of the energy density of the electromagnetic field is equal to minus the divergence of the field energy flux density.

# The Poynting's vector IV

This equation express, in differential form, the **work-energy theorem** of electrodynamics

It is to note that  $\mathbf{S}$  is a vector perpendicular to both  $\mathbf{E}$  and  $\mathbf{B}$ . Then the energy escape the volume  $V$  perpendicularly to the plane containing  $\mathbf{E}$  and  $\mathbf{B}$ .

We also note that the integral of  $\mathbf{S}$  over  $S$  goes to zero as  $r \rightarrow \infty$  for static electromagnetic fields. However when there are time-varying  $\mathbf{E}$  and  $\mathbf{B}$  fields it is found that they decay as  $r^{-1}$  then the integral can contribute with a finite term (remember that  $S$  goes as  $r^2$ ) to the rate of change of the energy of the system. This amount of energy flowing outward (or inward) at large distance is called *electromagnetic radiation*.

*Indeed, the Maxwell's equations predict the existence of electromagnetic waves carrying energy.*

# Monochromatic Plane Electromagnetic waves I

Maxwell's equations show a symmetry and an interdependence of  $\mathbf{E}$  and  $\mathbf{B}$ . This is better noticeable writing the equations for an empty space ( $\rho = 0$  and  $\mathbf{J} = 0$ ).

$$\nabla \cdot \mathbf{E}(t, \mathbf{r}) = 0 \quad (22)$$

$$\nabla \times \mathbf{E}(t, \mathbf{r}) = -\frac{\partial \mathbf{B}(t, \mathbf{r})}{\partial t} \quad (23)$$

$$\nabla \cdot \mathbf{B}(t, \mathbf{r}) = 0 \quad (24)$$

$$\nabla \times \mathbf{B}(t, \mathbf{r}) = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}(t, \mathbf{r})}{\partial t} \quad (25)$$

We note that a time-varying  $\mathbf{E}$  field generates a  $\mathbf{B}$  field which is everywhere perpendicular to the direction in which  $\mathbf{E}$  changes. A time varying  $\mathbf{B}$  field

# Monochromatic Plane Electromagnetic waves II

generates an  $\mathbf{E}$  field which is everywhere perpendicular to the direction in which  $\mathbf{B}$  changes.

One of the most important features of these equations is that they admit a solution in the form of a travelling wave carrying energy. Taking the curl of the second equation and combining it with the fourth:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad (26)$$

Taking into account the first equation we have:

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0 \quad (27)$$

An identical equation can be obtained for  $\mathbf{B}$



# Monochromatic Plane Electromagnetic waves III

$$\nabla^2 \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} = 0 \quad (28)$$

Equations 27 and 28 are “wave equations”.

We recall that a “wave” is a travelling disturbance carrying energy from place to place. Also waves are usually divided in *longitudinal wave*, when the disturbance caused by the wave moves along the direction that the wave propagates (e.g., sound waves), and *transverse wave*, the disturbance caused by the wave moves perpendicular to the direction that the wave propagates (e.g., water waves).

The existence of electromagnetic waves, predicted by Maxwell, was investigated by Heinrich Hertz. He, after experiments, succeeded in generating and detecting radio waves, which are sometimes called Hertzian waves in his honour.

# Monochromatic Plane Electromagnetic waves IV

The fundamental new feature of these equations is that electromagnetic waves **can propagate in vacuum**, all other kind of waves (longitudinal spring/mass waves, transverse waves on a string, longitudinal sound waves, etc.), needs a medium to propagate.

Also, electromagnetic waves propagate with a speed (*phase velocity*):

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c$$

Maxwell found that this velocity agreed with previous measurements of the speed of light, so he concluded that:

“This velocity is so nearly that of light [from work by Fizeau in 1849], that it seems we have strong reason to conclude that light itself (including radiant heat, and other radiations if any) is an electromagnetic disturbance in the form of waves propagated through the electromagnetic field”.

# Monochromatic Plane Electromagnetic waves V

Equations 27 and 28 are vector equations. So they are actually shorthand for three separate equations for each of the components.

The simplest solutions of these equations are the *harmonic plane waves*:

$$\mathbf{E}(t, \mathbf{r}) = \mathbf{E}_0 e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})} ; \quad \mathbf{B}(t, \mathbf{r}) = \mathbf{B}_0 e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})} \quad (29)$$

where  $\mathbf{k} = 2\pi/\lambda$  is the wave propagator or wave vector, and  $\lambda$  is the wavelength of the wave;  $\omega = 2\pi\nu$  is the angular frequency and  $\nu$  is the frequency. A critical difference with sound or other waves is that the coefficient of the wave  $\mathbf{E}_0$  is itself a vector. Thus, in addition to the direction  $\mathbf{k}$  the wave is moving, there is a direction  $\mathbf{E}_0 = \hat{\mathbf{e}}E_0$  describing which way the electric field is pointing. We call  $\hat{\mathbf{e}}$  the *polarization vector for the electric field*.

# Monochromatic Plane Electromagnetic waves VI

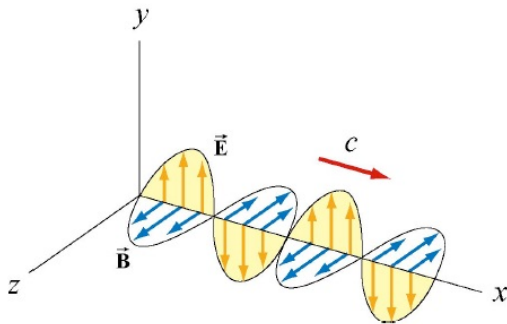


Figure:

# Monochromatic Plane Electromagnetic waves VII

**E** and **B** are measurable quantities and are defined as complex only for convenience but only the real part has a physical meaning. If we substitute **E** in equation 27 we have:

$$\nabla^2 \mathbf{E} = \mathbf{E}_0 \nabla^2 (e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})}) \quad (30)$$

$$= -k^2 \mathbf{E}_0 e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})} \quad (31)$$

$$= -k^2 \mathbf{E} \quad (32)$$

$$\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = -\omega^2 \mu_0 \epsilon_0 \mathbf{E} \quad (33)$$

we have:

$$k = \omega \sqrt{\mu_0 \epsilon_0} = \frac{\omega}{c} \quad (34)$$

# Monochromatic Plane Electromagnetic waves VIII

This equation, which relates the wave vector ( $k = 2\pi/\lambda$ ) and the frequency ( $\omega = 2\pi\nu$ ), is called the *dispersion relation*. The same relation can be found using the wave equation for **B**.

Applying the Gauss's law to the plane wave equation, we have.

$$\nabla \cdot \mathbf{E} = \mathbf{E}_0 \cdot \nabla e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})} = -i\mathbf{k} \cdot \mathbf{E}_0 e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})} = 0 \Rightarrow \mathbf{k} \cdot \mathbf{E}_0 \quad (35)$$

Then **E** is perpendicular to the direction of propagation of the wave.

An electromagnetic wave consists of **E** and **B** fields oscillating perpendicular to each other and to the direction of propagation (the same is true for **B**).

Thus, the electromagnetic waves are transverse waves.

Applying the Faraday-Maxwell equation we have:

# Monochromatic Plane Electromagnetic waves IX

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (36)$$

$$-i\mathbf{k} \times \mathbf{E} = -i\omega \mathbf{B} \quad (37)$$

Then:

$$\mathbf{k} \times \mathbf{E} = \omega \mathbf{B} \quad (38)$$

Thus,  $\mathbf{k}$ ,  $\mathbf{E}$ ,  $\mathbf{B}$  are mutually perpendicular and are in phase (i.e.,  $\mathbf{E}$  and  $\mathbf{B}$  reach the minimum (maximum) amplitude at the same time and at the same point along the direction of propagation).

Equation 38 is also saying that there is the following relation between the modules of  $\mathbf{E}$  and  $\mathbf{B}$ :

$$kE = \omega B \Rightarrow E = \frac{\omega}{k} B = cB \quad (39)$$

# Monochromatic Plane Electromagnetic waves X

We need to solve only a wave equation (for  $\mathbf{E}$  or  $\mathbf{B}$ ) and the other field is fixed by the above relation. Furthermore, the amplitude of the magnetic field is smaller by the factor  $c$  with respect to the electric field, so the effect of the magnetic field on observations is generally much smaller and often may be ignored in physical situations, though it is essential for the electric field to propagate.



## Energy flux of radiation I

Now we consider a plane electromagnetic wave propagating along the direction  $+x$ . We suppose that the electric component is given by  $\mathbf{E} = E_0 \cos(\omega t - kx) \hat{\mathbf{j}}$  and the magnetic component is given by  $\mathbf{B} = B_0 \cos(\omega t - kx) \hat{\mathbf{k}}$ , the energy flux of this wave is given by the Poynting vector:

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) = \frac{1}{\mu_0} E_0 B_0 \cos^2(\omega t - kx) \hat{\mathbf{i}} = \frac{1}{c\mu_0} E_0^2 \cos^2(\omega t - kx) \hat{\mathbf{i}} \quad [W/m^2] \quad (40)$$

As expected,  $\mathbf{S}$  points in the direction of wave propagation.  $\mathbf{S} = \mathbf{S}(t, x)$  oscillates in time, however, we take a time average, since this is what is measured.

The average value of  $\mathbf{S}$  is different from zero. Recalling that  $\cos^2(\alpha) = \frac{1 - \cos(2\alpha)}{2}$  we have:

# Energy flux of radiation II

$$\mathbf{S} = \frac{1}{c\mu_0} E_0^2 \left( \frac{1}{2} - \frac{1}{2} \cos[2(\omega t - kx)] \right) \hat{\mathbf{i}} \quad [W/m^2] \quad (41)$$

averaging over a cycle the average value of  $\cos[2(\omega t - kx)]$  goes to zero, we have :

$$\langle S \rangle = \frac{1}{2c\mu_0} E_0^2 \quad [W/m^2] \quad (42)$$

Therefore  $\langle S \rangle$  express the *Energy Flux* of the electromagnetic wave.

# Energy density, momentum and radiation pressure I

The stored energy density ( $u$ ) in the fields is:

$$u = \frac{1}{2}(\epsilon_0 \mathbf{E}^2 + \frac{1}{\mu_0} \mathbf{B}^2) = \epsilon_0 \mathbf{E}^2 = \epsilon_0 E_0^2 \cos^2(\omega t - kx) \quad (43)$$

then, averaging over a cycle:

$$\langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2 = c \epsilon_0 \mu_0 \langle S \rangle = \frac{\langle S \rangle}{c} [J/m^3] \quad (44)$$

Note that  $u$  has the units of  $J/m^3$ , which is analogous to pressure (N/m<sup>2</sup> or Pa). Then the average pressure exerted by the radiation is:

$$P_{rad} = \langle u \rangle = \frac{\langle S \rangle}{c} \quad (45)$$

## Energy density, momentum and radiation pressure II

This expression is valid for a total absorption of the radiation. In case of total reflection of the radiation at the surface, the above result must be multiplied by two.

We can also define the average momentum density of radiation noting that  $\langle u \rangle / c$  has the units of a momentum density ( $N \cdot s / m^3$ ):

$$\langle p \rangle = \frac{\langle u \rangle}{c} = \mu_0 \epsilon_0 \langle S \rangle \quad (46)$$

and then the momentum stored in the field is:

$$\mathbf{p} = \epsilon_0 \mu_0 \mathbf{S} = \epsilon_0 (\mathbf{E} \times \mathbf{B}) \quad (47)$$

Thus, even static fields have momentum if  $(\mathbf{E} \times \mathbf{B})$  is different from zero. Because the electromagnetic wave transports energy, it carries momentum  $\mathbf{p}$  and exerts a force  $\mathbf{F} = \frac{d\mathbf{p}}{dt}$  on any surface the wave encounters.

# Solar radiation pressure I

Historically, Kepler was the first to postulate, in 1619, the existence of a radiation pressure to explain why the tails of comets are always pointing away from the sun. We can see now that he was right!

The Energy flux per unit normal area at 1 AU ( $\approx 1.5 \cdot 10^{11}$  m) from the Sun is  $\langle S \rangle \approx 1367$  [W/m<sup>2</sup>] then the solar radiation pressure is  $P_{rad} \approx 1367/3 \cdot 10^8 = 4.5 \cdot 10^{-6}$  [N/m<sup>2</sup>].

Let us consider the forces on a  $1\mu$  diameter particle with a density of  $1$  g/cm<sup>3</sup>, as can be found in the tail of a comet. The projected area,  $A$ , of the particle is  $A = 7.8 \cdot 10^{-13}$  m<sup>2</sup>, and its mass,  $m = 5.2 \cdot 10^{-16}$  kg. The force exerted by sunlight on this particle is:

$$F = P_{rad}A \approx 3.6 \cdot 10^{-18} \text{ [N]} \quad (48)$$

Then the particle will have an acceleration:  $a = \frac{F}{m} \approx 6.8 \cdot 10^{-3}$  m/s<sup>2</sup>, that is greater than the acceleration due to the solar gravity,  $g_{\odot}(1 \text{ AU}) =$

## Solar radiation pressure II

$G \frac{M_{\odot}}{(1AU)^2} \approx 5.9 \cdot 10^{-3} \text{ m/s}^2$  at 1 AU. It can be demonstrated that the ratio  $a/g_{\odot}$  is constant independently from the distances from the Sun. This is why the tail of comets point away from the sun.

# Radiation from an accelerated charge (from Longair) I

Much of what we need to understand about radiation processes in astronomy is related to the physics of the radiation of accelerated charged particles. The central relation is the radiation loss rate of an accelerated (along a line) charged particle in the non-relativistic limit, the **Larmor's Formula**.

$$-\left(\frac{dE}{dt}\right)_{rad} = \frac{q^2 |\ddot{\mathbf{r}}|^2}{6\pi\epsilon_0 c^3} = \frac{|\ddot{\mathbf{p}}|^2}{6\pi\epsilon_0 c^3} \quad (49)$$

$\mathbf{p} = q\mathbf{r}$  is the dipole moment of the accelerated electron with respect to some origin. This formula is very closely related to the radiation rate of a dipole radio antenna and so is often referred to as the radiation loss rate for dipole radiation.

It is to note that:

$$-\left(\frac{dE}{dt}\right) = \frac{q^2}{6\pi\epsilon_0 c^3} |\ddot{\mathbf{r}}|^2 \propto \left(\frac{\mathbf{F}}{m}\right)^2$$

## Radiation from an accelerated charge (from Longair) II

because  $\ddot{\mathbf{r}} = \mathbf{F}/m$ . Then the power is in general inversely proportional to the square of the mass of the emitting particle  $\rightarrow$  **Electrons emit much more than protons.**

Often the Larmor formula is expressed in c.g.s units:

$$-\left(\frac{dE}{dt}\right) = \frac{2q^2}{3c^3} |\ddot{\mathbf{r}}|^2$$

The **cooling time** of the emission process is given by:

$$t_{cool} = \frac{E}{|dE/dt|}$$

If the particle is moving with a velocity  $v \sim c$  the power emitted is (Lienard, 1898):



# Radiation from an accelerated charge (from Longair) III

$$-\left(\frac{dE}{dt}\right) = \frac{q^2}{6\pi\epsilon_0 c^3} \gamma^6 \left( |\ddot{\mathbf{r}}|^2 - \frac{1}{c^2} |\mathbf{v} \times \dot{\mathbf{r}}|^2 \right)$$

where:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

is the Lorentz factor.

# The Spectrum of radiation I

For astrophysical applications, we wish to specify a spectrum of radiation that is how the power is distributed over frequency.

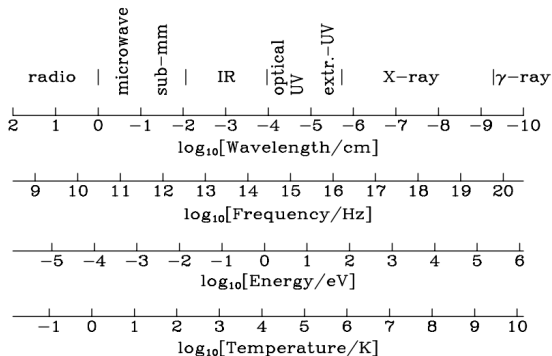


Figure: The electromagnetic spectrum

# The Spectrum of radiation II

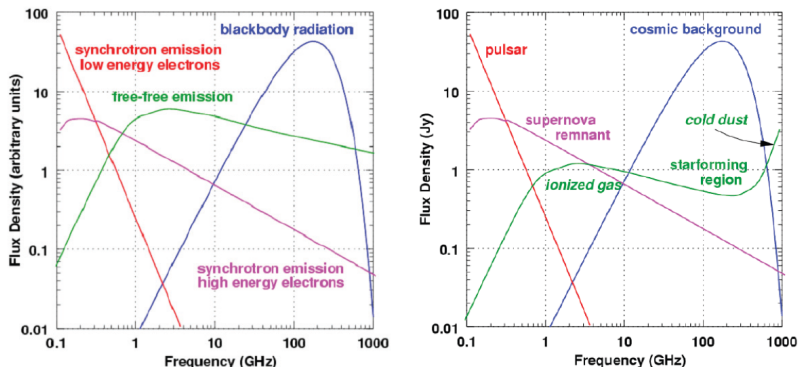


FIGURE 2.3 *Left:* Continuum spectra produced by various emission mechanisms. *Right:* Types of astronomical sources that produce the corresponding radio continuum emission. The flux levels shown correspond to the strongest known sources. A Jansky (Jy) is  $10^{-26} \text{ W m}^{-2} \text{ Hz}^{-1}$ .

# Backbody and Photons I

The first scientific study of thermal radiation began with the optical experiments of della Porta at the end of the 16th century. This experiment were then repeated and extended by Rumford and Davy and led to a better knowledge of the characteristics of thermal radiation. But was the work of Pictet, de Saussure and Prevost who in Geneva by 1790 had established the concept that all bodies radiate heat, the higher the temperature the greater the radiation such that when two bodies at different temperatures are within sight of each other the hotter will supply heat to the colder until they reach the same temperature. This result, known as Prevost's Theory of Exchanges was the starting point from which the modern physical theory of thermal radiation.

In 1800 William Herschel discovered the existence of a infrared radiation. Putting a thermometer behind the radiation corresponding to red light of the solar spectrum (obtained by decomposition through an optical prism), Herschel detected the presence of a radiation which, in spite of not being visible, was able to carry enough energy to increase the thermometer temperature.

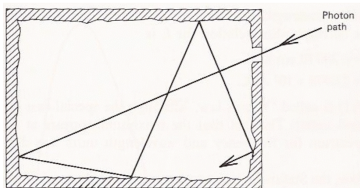
## Blackbody and Photons II

Kirchhoff make an attempt to summarize the state of knowledge in radiative heat transfer during the mid-1800's.

A **blackbody spectrum** emerges from a system in which matter and radiation are in **thermodynamic equilibrium**, which means that in a sufficiently small volume at any position there is a complete **thermal, mechanical, and chemical equilibrium**; that is **a single temperature is able to describe the state of the system**.

If we have a container that is completely closed except for a very small hole in one wall, any light entering the hole has a very small probability of finding its way out again, and will eventually be absorbed by the walls or the gas inside the container: this is a **perfect absorber**. Of course there is a very small chance that the photon will find the hole again and get out. This kind of container, or more specifically the hole is called a blackbody.

# Backbody and Photons III



If the container is heated, the walls emit photons filling the inside with radiation. Each photon is reabsorbed (the hole being negligible), that is a blackbody is also a **perfect emitter**.

Kirckhoff found that for a blackbody the spectrum of its radiation (depending only on its temperature) should be:

$$B_{\nu}(T) = \frac{j_{\nu}}{\kappa_{\nu}}$$

where  $j_{\nu}$  is the emission coefficient and  $\kappa_{\nu}$  is the absorption coefficient.

# Blackbody and Photons IV

The blackbody radiation is isotropic (the intensity is the same in all direction) but this is not in general the case. When we have a non-isotropic situation involving the flow of radiation from a preferred direction we need some mathematical prescription to describe it.

One of the most puzzling phenomena around 1900 was the to determine  $B_\nu(T)$  the function representing the spectral distribution of blackbody radiation.

In 1900 with an "act of despair" Planck introduced the hypothesis that matter can only emit and absorb discrete packets of electromagnetic energy :

$$E = h\nu \text{ [J]}$$

where  $\nu$  is the frequency of the radiation and  $h = 6.626 \cdot 10^{-34} \text{ [J} \cdot \text{s]}$  is the fundamental Planck constant. With this assumption Plank found the expression of the energy density of blackbody radiation, per frequency interval:

$$u_\nu(T) = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{h\nu/kT} - 1}$$

# Blackbody and Photons V

from this we can derive the form of  $B_\nu(T)$ , that is now known as Planck's function, the intensity of the radiation emitted by a blackbody having temperature  $T$ .

The famous Young's slit experiment is a demonstration, that light behaves like a wave.

However it is also possible to demonstrate that light behaves like a particle. When a metal plate is illuminated with blue or ultraviolet light, electrons absorb the energy from the light, and can escape from the metal. This phenomenon is known as the photoelectric effect. If light is a wave, we might think that the energy of the escaping electrons would increase as the intensity of light increased, but the frequency of that light would not matter. The strange thing was that the energy of the released electrons increases in proportion to the frequency of the light, and below a certain frequency, no electrons are emitted from the metal at all. *This cannot be explained by thinking about light as a wave.*

To solve this problem, in 1905, Einstein introduced the possibility that also the radiation field could be quantized, he wrote: "According to the assumption to be contemplated here, when a light ray is spreading out from a point, the energy is not



# Blackbody and Photons VI

distributed continuously over ever-increasing spaces, but consists of a finite number of energy quanta that are localized in points in space, move without dividing, and can be absorbed or generated only as a whole."

Having this in mind, he suggested that the photons carry energy and momentum, but not mass. The energy of the a photon is:

$$E = h\nu$$

An electron requires a certain amount of energy to free it from the metal. If we call this amount of energy  $W$  (work function) the energy of the freed electron should be given by  $E = h\nu - W$ . Thus, the energy of the emerging electron is proportional to the frequency of the incident light, just as we see in the photoelectric effect.

If the photon is moving in the direction  $\mathbf{n}$  its momentum is:

$$\mathbf{p} = \frac{h\nu}{c} \mathbf{n} = \frac{h}{\lambda} \mathbf{n} = \frac{h}{2\pi} k \mathbf{n} = \hbar k \mathbf{n}$$

## Backbody and Photons VII

The term *photon* used to indicate this energy quanta, was introduced, in 1926, by Gilbert N. Lewis, professor of theoretical chemistry in Berkeley. At high frequency the energy of the photon is often expressed in [eV]:

$$1\text{eV} = 1.6 \cdot 10^{-19}\text{J} \implies 1.6 \cdot 10^{-12}\text{erg}$$

$$E = \frac{6.626 \cdot 10^{-34}}{1.6 \cdot 10^{-19}} \nu = 4.1 \cdot 10^{-15} \nu \text{ [eV]}$$

If a photon has a frequency of  $\sim 2.42 \cdot 10^{14} \text{ Hz}$  its energy is 1 eV.

Assuming a condition in which matter and radiation are in thermal equilibrium:

$$E = h\nu = kT \implies T = \frac{h\nu}{k} = \frac{6.626 \cdot 10^{-34}}{1.38 \cdot 10^{-23}} \nu = 4.8 \cdot 10^{-11} \nu \text{ [K]}$$

If the energy of the photons is 1 eV the associated temperature is  $T \sim 10000 \text{ K}$ .

# Radiation Theory I

With several exceptions, astronomical phenomena are almost always observed by detecting and measuring electromagnetic radiation, that is the superposition of waves having different frequencies, phases and polarization directions (*incoherent radiation*) emitted by distant sources.

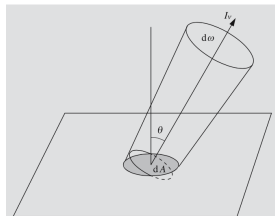
We now introduce the main quantities that are used to describe a generic radiation field.

# Radiation Theory II

Let assume to have a generic radiation field, if we place a surface element  $dA$ , with normal  $\mathbf{n}$ , in this field, some radiation will pass through  $dA$ . Some of the radiation will leave  $dA$  within a solid angle  $d\omega$  ( $d\omega = \sin\theta d\theta d\varphi = -d(\cos\theta)d\varphi$ ) around the direction of interest that form an angle  $\theta$  with the normal to the surface element. The amount of energy with frequency in the range  $(\nu, \nu + d\nu)$  entering this solid angle in time  $dt$  is:

$$dE_\nu = I_\nu(\theta, \varphi) \cos\theta dA d\nu d\omega dt$$

the coefficient  $I_\nu(\theta, \varphi)$  is the **radiation (specific) intensity** at the frequency  $\nu$  in the direction of the solid angle  $d\omega$ . Its dimensions is  $Wm^{-2}Hz^{-1}sterad^{-1}$  or  $ergs^{-1}cm^{-2}Hz^{-1}sterad^{-1}$  (c.g.s.)



# Radiation Theory III

We can relate the Intensity to the wavelength instead of frequency. Because the total amount of energy remain the same:

$$I_\nu d\nu = -I_\lambda d\lambda$$

Because  $\nu = c/\lambda$  we have  $d\nu = -\frac{c}{\lambda^2} d\lambda$  then:

$$I_\lambda = \frac{c}{\lambda^2} I_\nu$$

The intensity including all possible frequencies is called the **total intensity**  $I$ , and is given by:

$$I(\theta, \varphi) = \int_0^\infty I_\nu(\theta, \varphi) d\nu$$

We also define the **mean intensity** averaged over all the solid angles:

$$J_\nu = \frac{1}{4\pi} \int_\Omega I_\nu(\theta, \varphi) d\omega$$

# Radiation Theory IV

If the radiation field is isotropic  $J_\nu = I_\nu$

The **average intensity** of the total intensity is:

$$J = \int_0^\infty J_\nu d\nu$$

More important quantities from the observational point of view are the **energy flux** or the **flux** and the **flux density**.

The flux density  $F_\nu$  at a frequency  $\nu$  can be expressed in terms of the intensity as:

$$F_\nu = \frac{1}{dAd\nu dt} \int_\Omega dE_\nu = \int_\Omega I_\nu(\theta, \varphi) \cos\theta d\omega \quad [Wm^{-2}Hz^{-1}]$$

where the integration is extended over all possible directions. For a narrow ray this is the Poynting's flux in e.m. The flux density gives the power of radiation per unit area. The **total flux density** is:

$$F = \int F_\nu d\nu \quad [Wm^{-2}]$$

# Radiation Theory V

If the radiation is isotropic, i. e. if  $I$  is independent of the direction, we get:

$$F = \int_{\Omega} I \cos\theta d\omega = I \int_{\Omega} \cos\theta d\omega$$

The solid angle element  $d\omega$  is equal to a surface element on a unit sphere:  $d\omega = \sin\theta d\theta d\phi$ . Then

$$F = I \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \cos\theta \sin\theta d\theta d\phi = 0$$

so there is no net flux of radiation. This means that there are equal amounts of radiation entering and leaving the surface. the amount of radiation passing through the surface is:

$$F = I \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} \cos\theta \sin\theta d\theta d\phi = \pi I$$

# Radiation Theory VI

The total flux of a star is called the **luminosity**  $L$  and is given by (specific luminosity):

$$L_\nu = 4\pi R^2 F_\nu \quad [W]$$

and (total Luminosity)

$$L = 4\pi R^2 F \quad [W]$$

This expression is valid for a star having a radius  $R$  and emitting isotropically . If we are outside the source, where radiation is not created or destroyed, the luminosity does not depend on distance. The flux density, on the other hand, falls off proportional to  $1/r^2$  (the *inverse square law* of light).

The **observed flux** at a distance  $d$  from the source is:

$$f_\nu = \frac{L_\nu}{4\pi d^2} = \frac{R^2}{d^2} F_\nu$$

The **energy density**  $u$  of radiation is the amount of energy per unit volume. To derive  $u$  let suppose we have radiation with intensity  $I$  arriving from a solid angle



# Radiation Theory VII

$d\omega$  perpendicular to the surface  $dA$ . In the time  $dt$ , the radiation travels a distance  $c dt$  and fills a volume  $dV = c dt dA$ .

Thus the energy in the volume  $dV$  is (with  $\cos\theta = 1$ ):

$$dE = I dA d\omega dt = \frac{1}{c} I d\omega dV$$

then

$$du = \frac{dE}{dV} = \frac{1}{c} I d\omega$$

and the total energy density is obtained by integrating this over all directions:

$$u = \frac{1}{c} \int_{\Omega} I d\omega = \frac{4\pi}{c} J \quad [Jm^{-3}]$$

if we consider only one frequency:

$$u_{\nu} = \frac{4\pi}{c} J_{\nu}$$

# Radiation Theory VIII

In terms of photon density:

$$n_\nu = \frac{u_\nu}{h\nu} = \frac{4\pi}{c} \frac{J_\nu}{h\nu}$$

Every beam of radiation transports not only energy in the form of flux  $F_\nu$ , but also a momentum. Thus the radiation incident on a surface applies a **radiation pressure** on that surface.

Consider photons bouncing back and forth between two perfect mirrors having a surface  $A$  and separation  $L$ . Each photon has energy  $E_\nu = h\nu$  and momentum  $p_\nu = E_\nu/c$  and its contribution to the energy density is  $\Delta u_\nu = E_\nu/(AL)$ . For each photon incidents on one of the two mirrors with an angle  $\theta$  with respect to the normal to the mirror, there is a transfer of momentum to the mirror given by:

$$\Delta p_\nu = p_{\nu,ref} - p_{\nu,inc} = \frac{2}{c} E_\nu \cos \theta$$

The time interval between bounces is:

# Radiation Theory IX

$$\Delta t = \frac{2L}{c \cos \theta}$$

Then the force exerted by the photon on the mirror is:

$$\frac{\Delta p_\nu}{\Delta t} = \frac{E_\nu}{L} \cos^2 \theta$$

and the contribution to the radiation pressure is:

$$\Delta P_{r,\nu} = \frac{\Delta p_\nu}{A \Delta t} = \frac{E_\nu}{AL} \cos^2 \theta = \Delta u_\nu \cos^2 \theta$$

If the radiation is isotropic to obtain the total pressure we have to integrate over all the direction, this is equivalent to evaluate the average value of  $\cos^2 \theta$  over a sphere,  $\langle \cos^2 \theta \rangle = 1/3$ , then recalling that  $u_\nu = (4\pi/c)J_\nu$

$$P_{r,\nu} = \frac{u_\nu}{3} = \frac{4\pi}{3c} J_\nu$$

# Radiation Theory X

Integrating over all frequency:

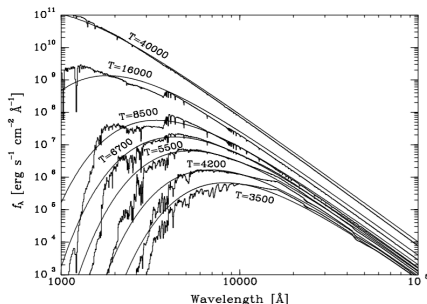
$$P_r = \frac{u}{3} = \frac{4\pi}{3c} J$$

It can be demonstrated that the pressure is equal to 1/3 times the energy density, is valid non only for photons, but also for relativistic gas, in which the velocity of the particles is close to  $c$ .

# Blackbody radiation properties I

There are several specific processes that can produce radiation. These processes can produce different spectra that is different  $f_\nu$ .

For example, to a very rough, but quite useful, approximation, stars shine with the spectrum of a **blackbody**.



# Blackbody radiation properties II

The Intensity of the Blackbody radiation is

$$I_\nu = c \frac{du_\nu}{d\omega}$$

Blackbody radiation is isotropic and hence:

$$\frac{du_\nu}{d\omega} = \frac{u_\nu}{4\pi}$$

Then

$$I_\nu \equiv B_\nu(T) = \frac{c}{4\pi} u_\nu = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1}$$

In terms of  $\lambda$ :

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/k\lambda T} - 1}$$

We can consider two important limit cases:

# Blackbody radiation properties III

$\frac{h\nu}{kT} \gg 1$  : in this limit we can drop the term unity in the denominator and then obtain the **Wien's Law**:

$$B_\nu(T) \simeq \frac{2h\nu^3}{c^2} e^{-h\nu/kT}$$

$\frac{h\nu}{kT} \ll 1$  : in this limit we can expand the exponential:  $e^{\frac{h\nu}{kT}} - 1 = \frac{h\nu}{kT} + \dots$  considering only the first order term of the expansion. This leads to the **Rayleigh-Jeans Law**:

$$B_\nu(T) \simeq \frac{2h\nu^2 kT}{c^2}$$

# Blackbody radiation properties IV

The flux density  $F_\nu$  of a blackbody is:

$$F_\nu = \pi B_\nu(T) = \frac{c}{4} u_\nu = \frac{2\pi h \nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1}$$

In terms of  $\lambda$ :

$$F_\lambda = \frac{2\pi h c^2}{\lambda^5} \frac{1}{e^{hc/k\lambda T} - 1}$$



# Blackbody radiation properties V

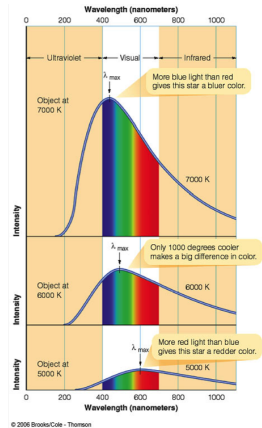
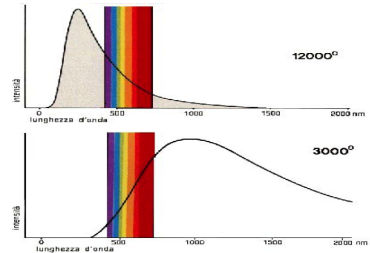


Figure: The blackbody spectrum for different values of  $T$ .

# Blackbody radiation properties VI

The **Stefan-Boltzmann Law** gives the total radiation flux emitted by a blackbody unit surface inside the cavity:

$$F = \frac{c}{4} \int_0^{\infty} u_{\nu} d\nu = \frac{c}{4} a T^4 = \sigma T^4$$

where:

$$a = 7.6 \times 10^{-15} \text{ erg cm}^{-3} \text{ K}^{-4}$$

and

$$\sigma = \frac{2\pi^5 k^4}{15c^2 h^3} = 5.67 \times 10^{-5} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ K}^{-4}$$

or:

$$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

Then if we assimilate the star to a backbody having a temperature  $T$  and radius  $R$ , its luminosity is:

$$L = 4\pi R^2 F = 4\pi R^2 \sigma T^4$$

# Blackbody radiation properties VII

Thus observed flux at a distance  $d$  from the source is:

$$f = \frac{L}{4\pi d^2} = \frac{R^2}{d^2} \sigma T^4$$

The wavelength or frequency of the peak of a blackbody spectrum can be found by taking its derivative and equating to zero:

$$\frac{dB_\nu}{d\nu} = 0$$

which lead to the **Wien's law**:

$$h\nu_{max} = 2.8 kT$$

or expressed in terms of  $\lambda$ :

$$\lambda_{max} T = 0.29 \text{ cm K}$$

# Blackbody radiation properties VIII

For example, **the Sun** which radiates approximately like a blackbody at  $T = 5800K$ , maximum emission is at  $\lambda = 500nm$ . which is the wavelength of green light, in the middle of the visual regime.

# Blackbody radiation properties IX

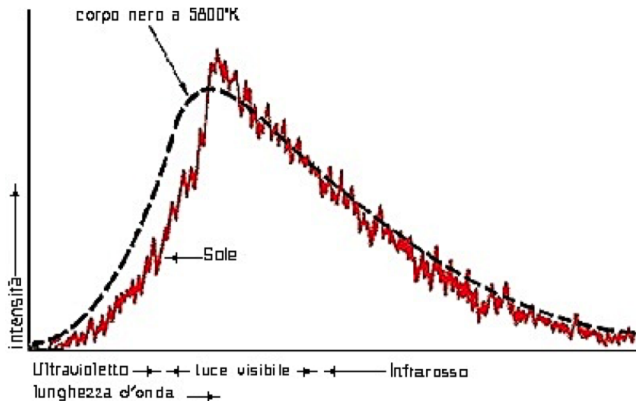
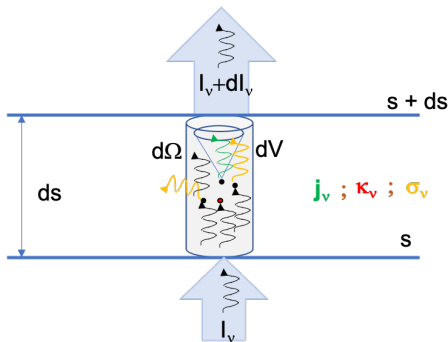


Figure: Solar Spectrum

# Emission and Absorption & Transport Equation I

If a beam of radiation traverses a layer of matter, energy can be added or subtracted from it by emission, absorption or scattering of photons into and out of the beam.



# Emission and Absorption & Transport Equation II

## Emission:

The spontaneous monochromatic emission coefficient  $j_\nu$  is defined as the energy emitted per unit of time per unit solid angle per unit volume in the frequency range  $\nu, \nu + d\nu$ :

$$dE = j_\nu dV d\Omega dt d\nu$$

where  $j_\nu$  has unit of  $W m^{-3} sr^{-1} Hz^{-1}$ . When the beam crosses a distance  $ds$  if its cross section is  $dA$ , it travels through a volume  $dV = dA ds$ , thus the intensity added to the beam by spontaneous emission is:

$$dl_\nu = j_\nu ds$$

## Absorption:

# Emission and Absorption & Transport Equation III

The amount of absorption that the beam of radiation experiences in passing through a layer of matter of thickness  $ds$  is expressed by an absorption coefficient  $\kappa_\nu$  (having unit of  $m^{-1}$ ) in the equation:

$$dl_\nu = -\kappa_\nu l_\nu ds$$

in the general case  $\kappa_\nu$  includes also the amount of radiation removed from the beam by scattering ( $\kappa_\nu = \kappa_{\nu,a} + \kappa_{\nu,s}$ ).

## Transport equation:

We are now able to express the changes in the intensity of the radiation along the path  $ds$ :

$$\frac{dl_\nu}{ds} = -\kappa_\nu l_\nu + j_\nu$$



# Emission and Absorption & Transport Equation IV

The coefficients  $j_\nu$  and  $\kappa_\nu$ , depend on the type and state of the matter (chemical composition, pressure, temperature) but also on the direction of propagation of the beam in the material. It is convenient to introduce a dimensionless quantity called **optical thickness**:

$$\tau = \int_0^L \kappa_\nu ds$$

In terms of  $\tau$  the transport equation becomes ( $d\tau = \kappa_\nu ds$ ):

$$\frac{dl_\nu}{d\tau} = -l_\nu + \frac{j_\nu}{\kappa_\nu}$$

if we define a **Source function**:

$$S_\nu = \frac{j_\nu}{\kappa_\nu}$$

# Emission and Absorption & Transport Equation V

we have:

$$\frac{dl_\nu}{d\tau} = -I_\nu + S_\nu$$